

EE4730TU High Frequency Wireless Architectures

Project2: Antenna measurements at high frequencies. Far-field and Near-Field

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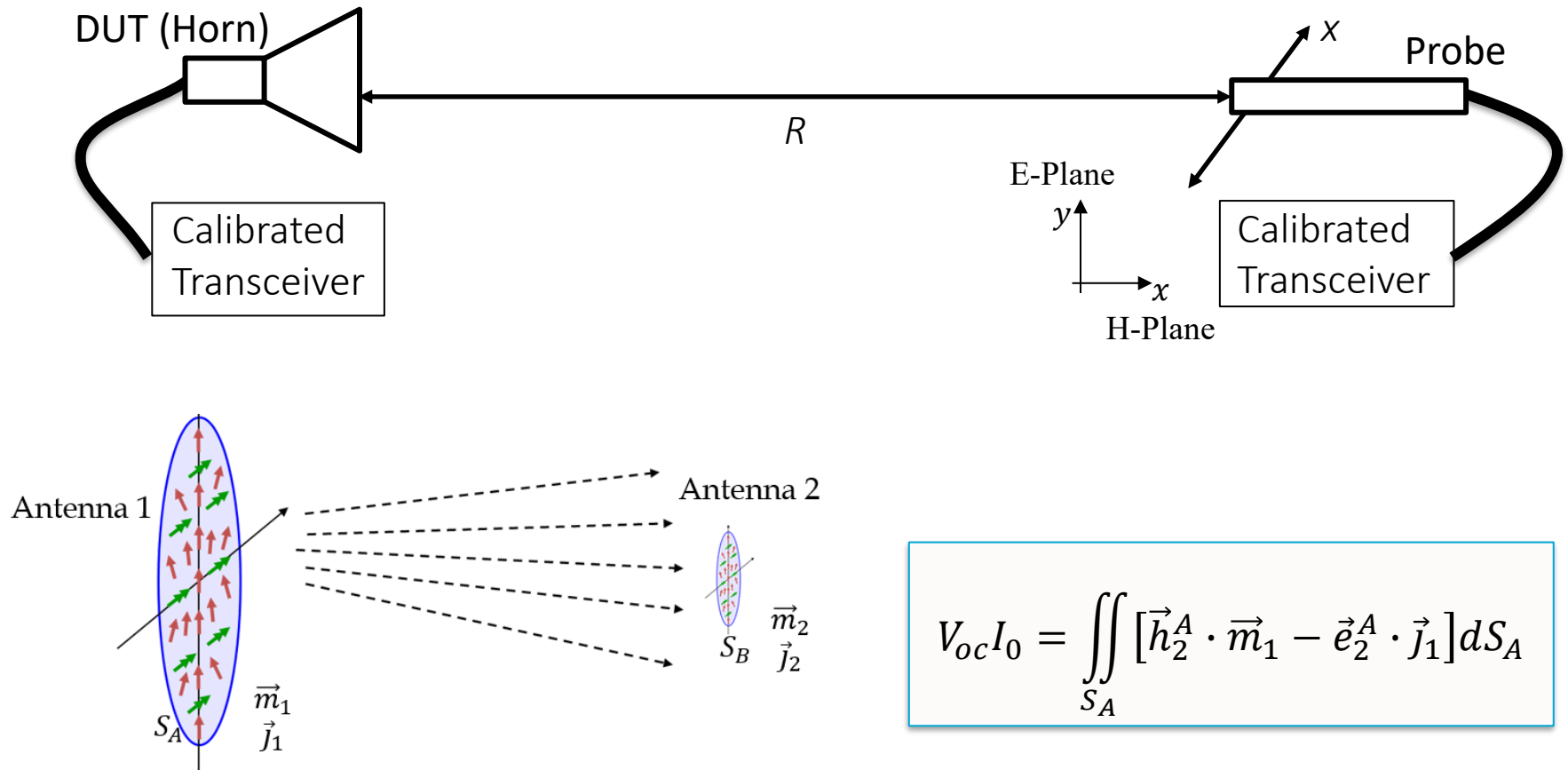
Mekelweg 4, 2628CD Delft

<http://terahertz.tudelft.nl>

Project #2 – Evaluate the measured antennas

- Evaluate antennas in the far-field and near-field.
 - Coupling
 - Radiation Pattern
 - Directivity
 - Gain
- Apply time-gating techniques to reduce the multiple reflections of the environment

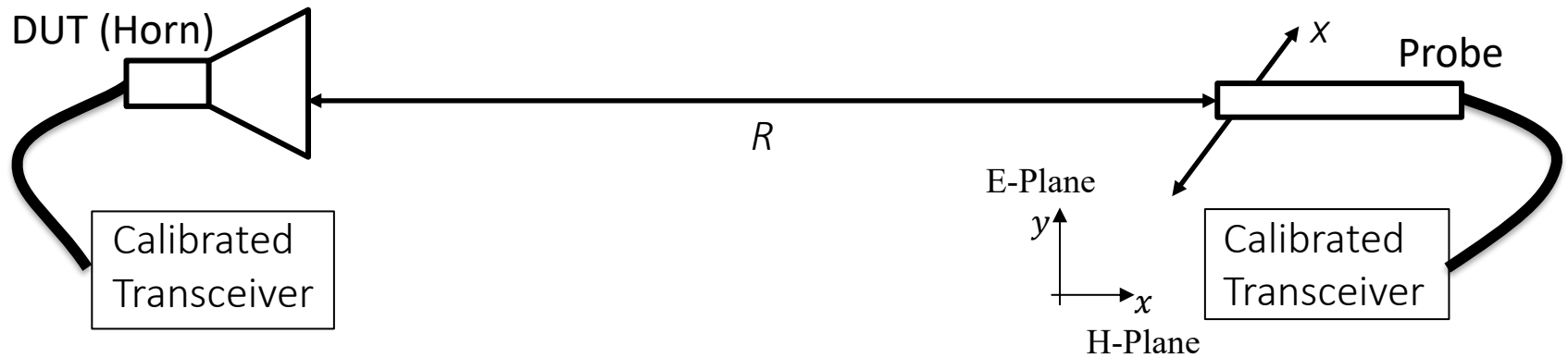
Measurements in the far-field using a planar scanner



$$V_{oc}I_0 = \iint_{S_A} [\vec{h}_2^A \cdot \vec{m}_1 - \vec{e}_2^A \cdot \vec{j}_1] dS_A$$

$$P_{rx} = \frac{|V_{oc}I_0|^2}{16P_{rad}^2} \quad \longrightarrow \quad Cou = \frac{P_{rx}}{P_{rad}^1} \quad \longrightarrow \quad Cou = \frac{|V_{oc}I_0|^2}{16P_{rad}^2 P_{rad}^1}$$

Measurements in the far-field using a planar scanner

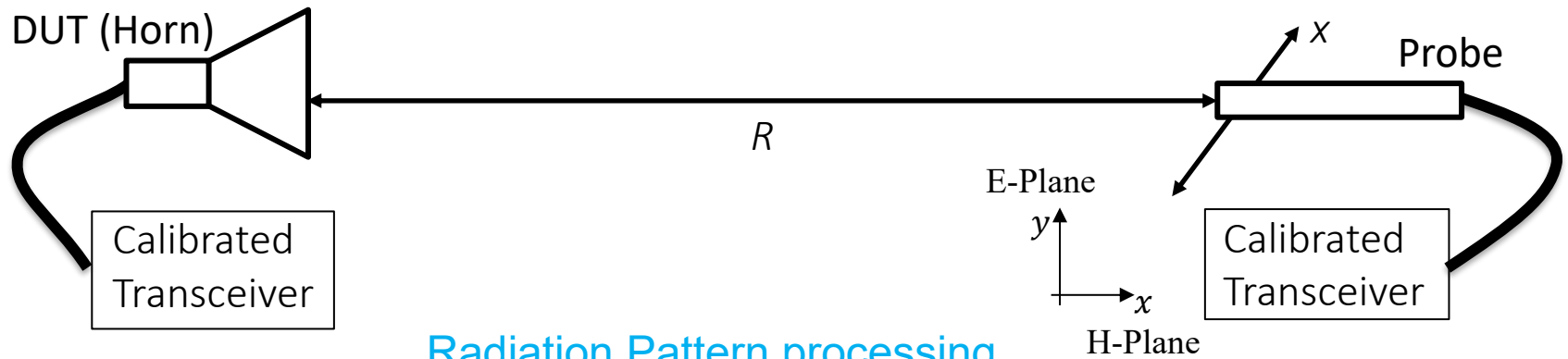


$$Cou \approx \frac{k^2 A_2}{A_1 R^2} \left| \frac{A_1}{2\pi} \right|^2 = \frac{k^2 A_2 A_1}{4\pi^2 R^2} = \frac{A_2 A_1}{\lambda^2 R^2}$$

Friis Equation:

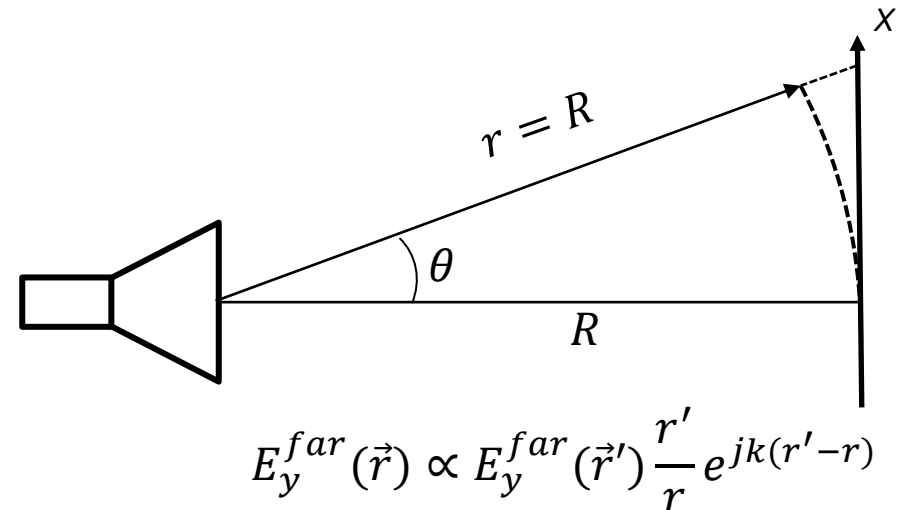
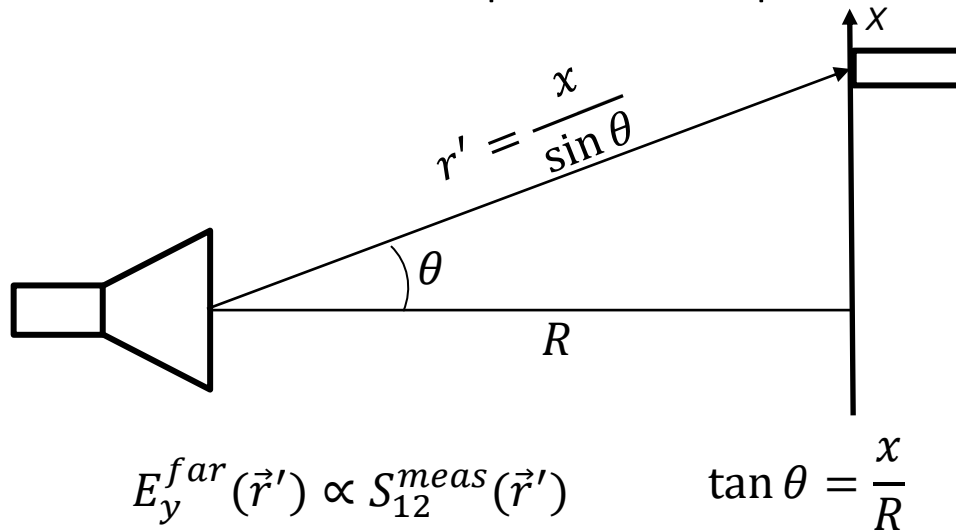
$$\frac{P_{rx}}{P_{tx}} = Cou = G_{tx} G_{rx} \left(\frac{\lambda}{4\pi R} \right)^2 \quad \Rightarrow \quad Cou = \frac{G_{tx}}{4\pi R^2} A_{ef}^{rx} \quad \Rightarrow \quad Cou = \frac{A_{ef}^{tx} A_{ef}^{rx}}{\lambda^2 R^2}$$

Measurements in the far-field using a planar scanner



Radiation Pattern processing

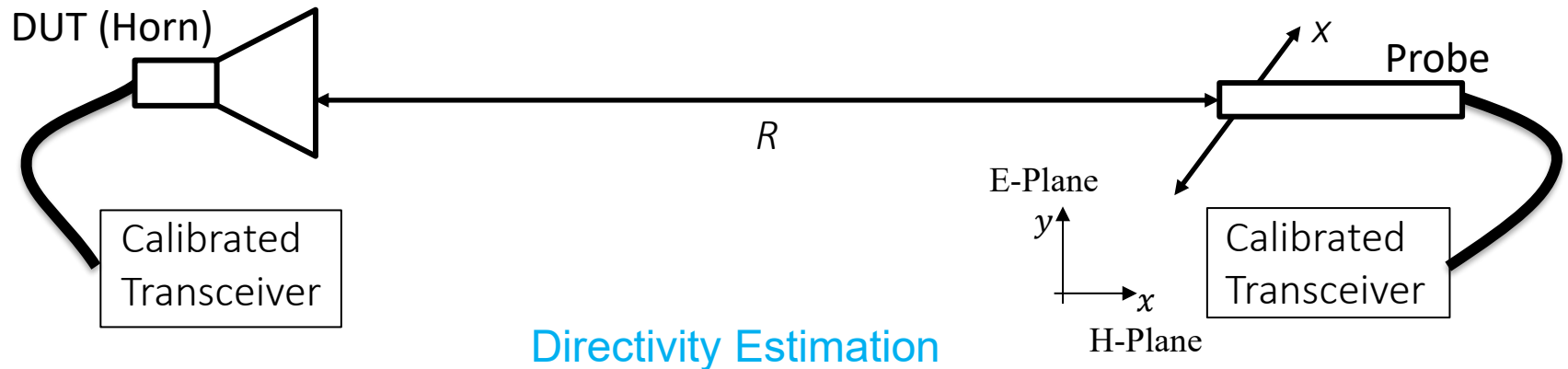
1. Convert from planar to spherical coordinates



2. Compensation for the probe radiation pattern

$$E_y^{DUT}(\vec{r}) \propto \frac{E_y^{far}(\vec{r})}{E_{probe}(\vec{r})}$$

Measurements in the far-field using a planar scanner



Directivity Estimation

3. Approximation of directivity assuming symmetry in ϕ

$$D = \frac{U_{max}}{P_{rad}/4\pi} = \frac{4\pi |E_y^{far}(\vec{r})|^2|_{max}}{\pi \int_{-\pi}^{\pi} |E_y^{far}(\vec{r})|^2 \sin \theta d\theta}$$

Gain Estimation

4. Gain (far-field)

$$\frac{P_{rx}}{P_{tx}} = G_{tx} G_{rx} \left(\frac{\lambda}{4\pi R} \right)^2$$

$$|S_{12}|^2 \approx \frac{P_{rx}}{P_{in}(1 - |S_{22}^{probe}|^2)}$$

- Horn to Horn
- Horn to Probe

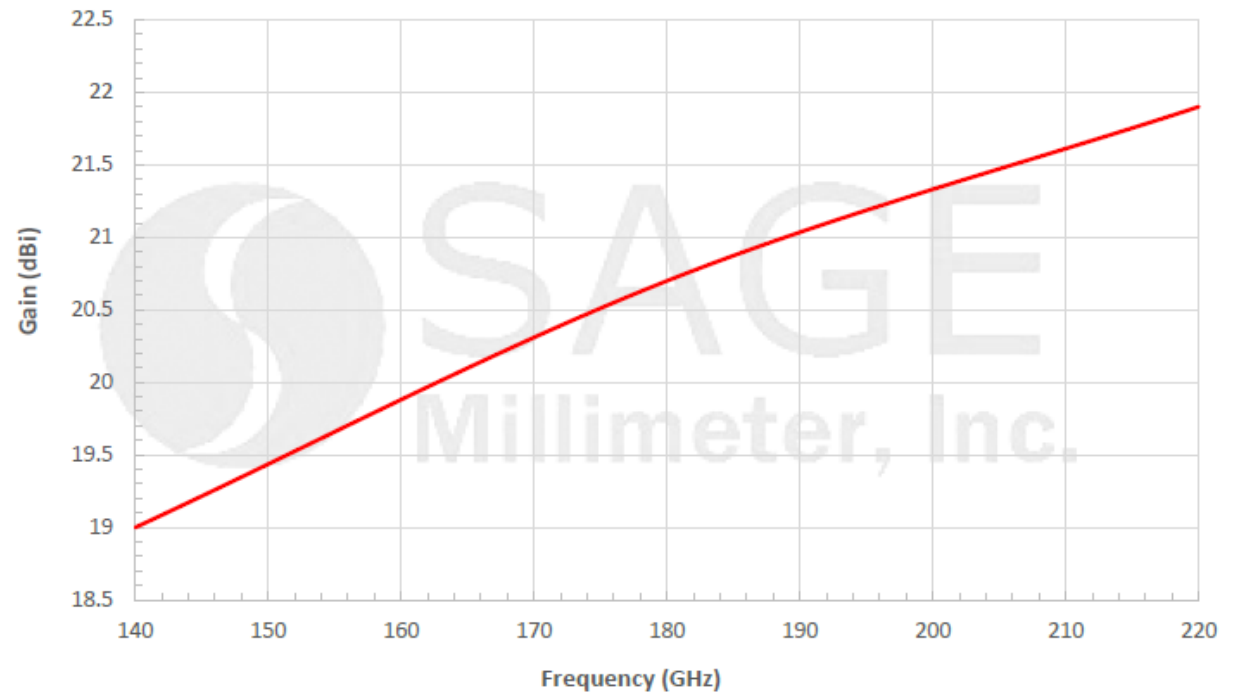
Pyramidal Horn Antenna 20 dBi



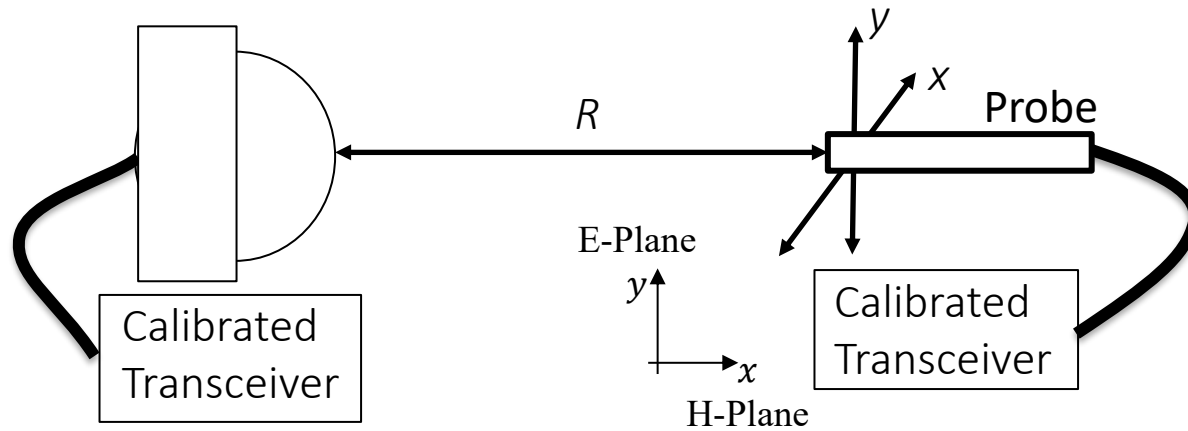
SAR-2013-05-S2

WR-05 Pyramidal Horn Antenna, 20 dBi Gain

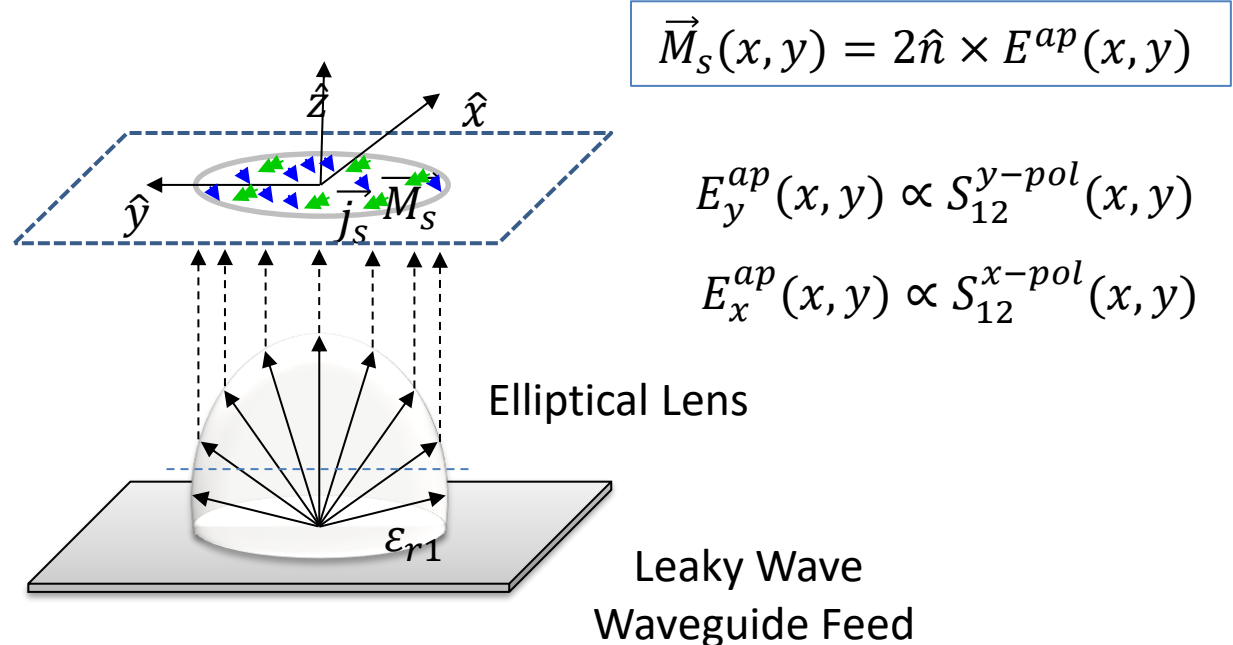
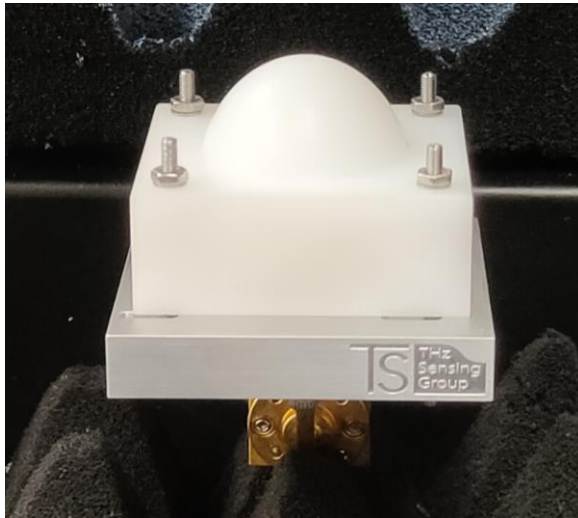
Typical Gain vs. Frequency



Measurements in the near-field using a planar scanner



1. Convert from near-field to far-field



Measurements in the near-field using a planar scanner

$$\vec{M}_s(\vec{r}') = -2\hat{z} \times (E_y^{ap}(\vec{r}')\hat{y} + E_x^{ap}(\vec{r}')\hat{x}) = 2(S_{12}^y(\vec{r}')\hat{x} - S_{12}^x(\vec{r}')\hat{y})$$

Field radiated by an antenna in the radiative near field:

$$\vec{E}^{far}(\vec{r}) \approx k\hat{r} \times \tilde{I} \iiint_v \vec{M}_s(\vec{r}') e^{jk\hat{r} \cdot \vec{r}'} d\vec{r}' \frac{je^{-jkr}}{4\pi r}$$

$$\begin{aligned} \vec{E}^{far}(\vec{r}) &\propto k\hat{r} \times \tilde{I} \iint_S 2(S_{12}^y(\vec{r}')\hat{x} - S_{12}^x(\vec{r}')\hat{y}) e^{jk\hat{r} \cdot \vec{r}'} d\vec{r}' \frac{je^{-jkr}}{4\pi r} \\ \vec{E}^{far}(\vec{r}) &\propto 2k \left(\begin{bmatrix} 0 \\ \sin \phi \\ \cos \theta \cos \phi \end{bmatrix} \iint_v S_{12}^x(\vec{r}') e^{jk\hat{r} \cdot \vec{r}'} d\vec{r}' - \begin{bmatrix} 0 \\ -\cos \phi \\ \cos \theta \sin \phi \end{bmatrix} \iint_v S_{12}^y(\vec{r}') e^{jk\hat{r} \cdot \vec{r}'} d\vec{r}' \right) \frac{je^{-jkr}}{4\pi r} \end{aligned}$$

$$\vec{E}^{far}(\vec{r}) \propto j2k \left([\tilde{S}_{12}^x(\theta, \phi) \sin(\phi) + \tilde{S}_{12}^y(\theta, \phi) \cos(\phi)] \hat{\theta} + [\tilde{S}_{12}^x(\theta, \phi) \cos(\theta) \cos(\phi) - \tilde{S}_{12}^y(\theta, \phi) \cos(\theta) \sin(\phi)] \hat{\phi} \right) \frac{e^{-jkr}}{4\pi r}$$

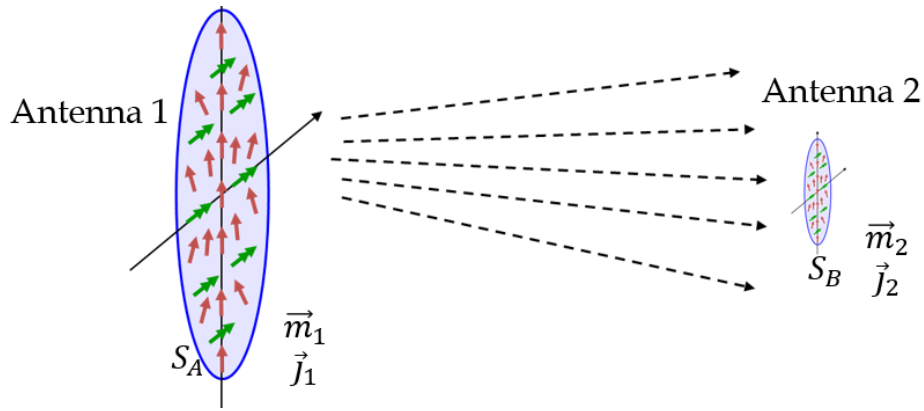
$$\tilde{S}_{12}^x(\theta, \phi) \propto \iint_S S_{12}^x(\vec{r}') e^{jk\hat{r} \cdot \vec{r}'} d\vec{r}' = \iint_S S_{12}^x(x, y) e^{j(k_x x + k_y y)} dx dy$$

$$\begin{aligned} k_x &= k \sin \theta \cos \phi \\ k_y &= k \sin \theta \sin \phi \end{aligned}$$

2. After that we would still need to compensate for the probe

$$E^{DUT}(\vec{r}) \propto \frac{E^{far}(\vec{r})}{E_{probe}(\vec{r})}$$

Power Coupling Coefficient



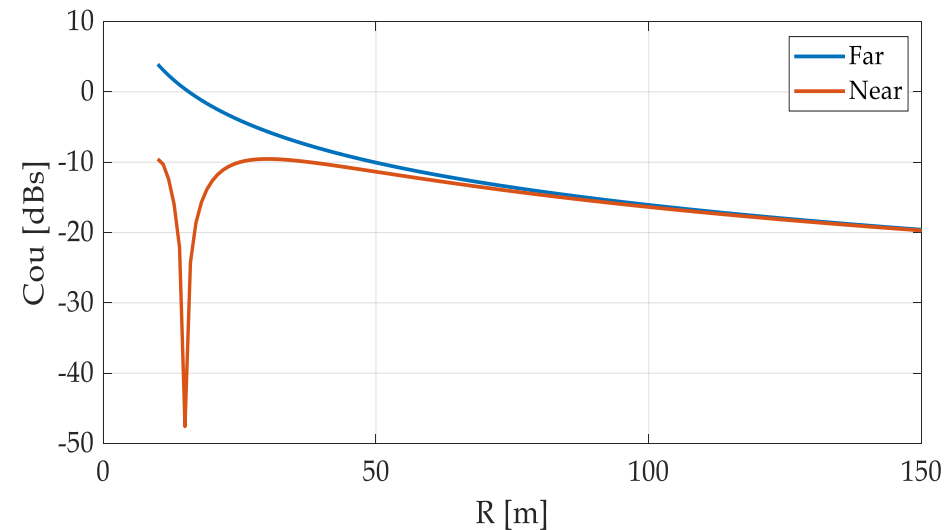
$$V_{oc}I_0 = \iint_{S_A} [\vec{h}_2^A \cdot \vec{m}_1 - \vec{e}_2^A \cdot \vec{j}_1] dS_A$$

$$P_{rx} = \frac{|V_{oc}I_0|^2}{16P_{rad}^2} \quad \longrightarrow \quad Cou = \frac{P_{rx}}{P_{rad}^1} \quad \longrightarrow \quad Cou = \frac{|V_{oc}I_0|^2}{16P_{rad}^2 P_{rad}^1}$$

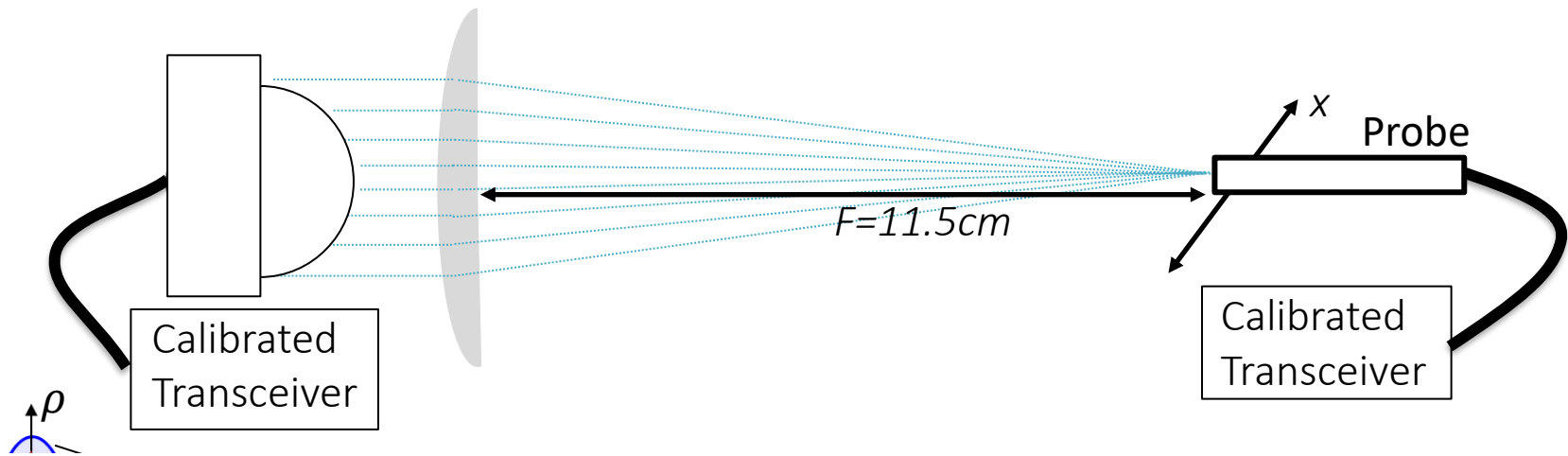
Small angle approx. and uniform aperture distribution:

$$V_{oc}I_0 \approx \frac{2jk\zeta A_2 e^{-jkR}}{R} \int_0^{D_1/2} e^{-jk\frac{\rho'^2}{2R}} \rho' d\rho'$$

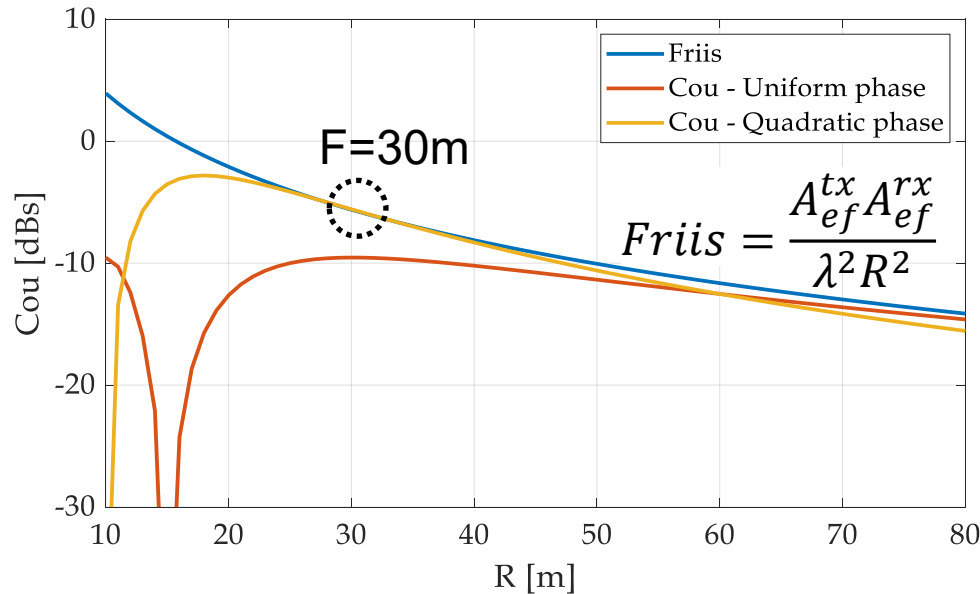
$$Cou = \frac{k^2 A_2^2}{A_1 R^2} \left| \int_0^{D_1/2} e^{-jk\frac{\rho'^2}{2R}} \rho' d\rho' \right|^2$$



Measurements in the near-field using a planar scanner



x



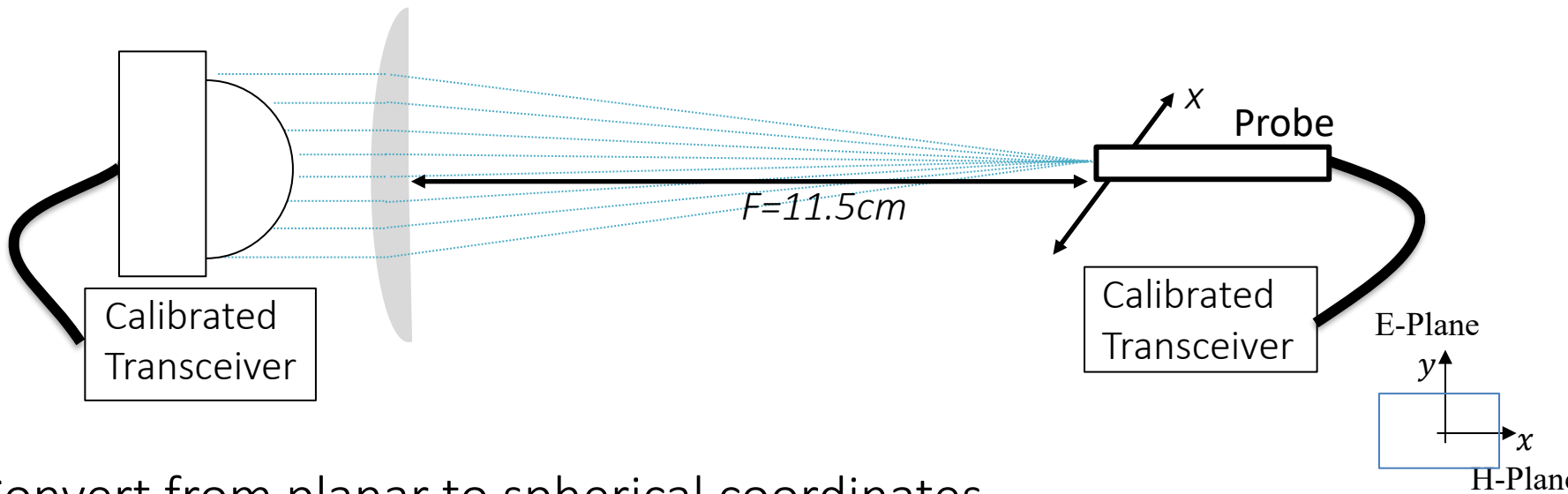
$$\vec{e}_{A1}(\rho_f) = \frac{jke^{-jkF}}{2\pi F} \int \vec{e}_2^{GO}(\vec{r}') e^{jk\vec{r}_f \cdot \hat{r}'} d\vec{r}'$$

$$\approx -\zeta \frac{jke^{-jkF}}{2\pi F} \text{Airy}\left(\frac{k}{F}\rho_f\right) \hat{y}$$

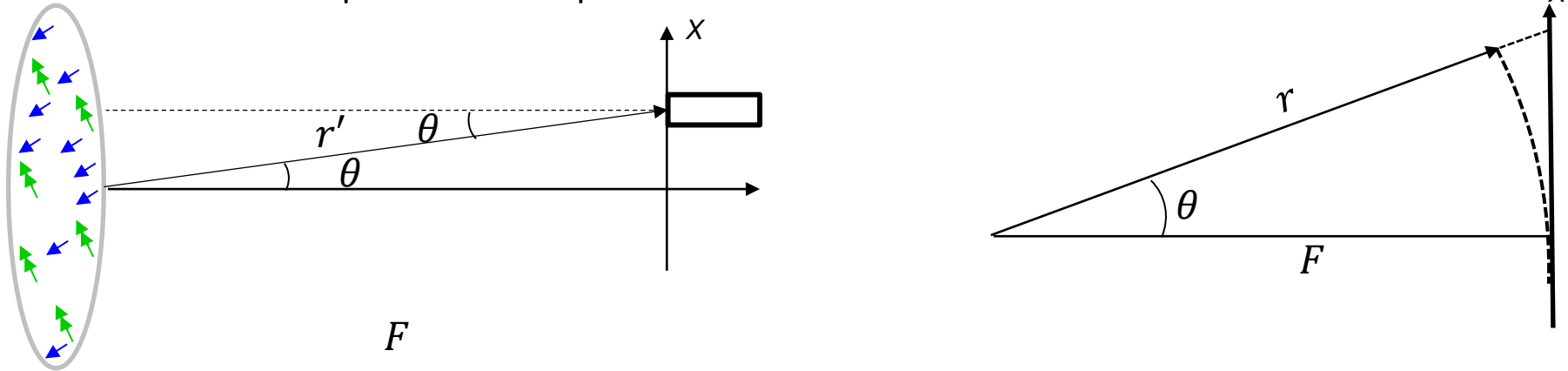
$$\text{Airy}\left(\frac{k}{F}\rho_f\right) = 2\pi \frac{D^2}{4} \frac{J_1\left(\frac{k}{F}\rho_f \frac{D}{2}\right)}{\frac{k}{F}\rho_f \frac{D}{2}}$$

$$Cou = \frac{k^2 A_2}{A_1 R^2} \left| \int_0^{D_1/2} e^{jk\frac{\rho'^2}{2F}} e^{-jk\frac{\rho'^2}{2R}} \rho' d\rho' \right|^2$$

Measurements in the near-field using a planar scanner



1. Convert from planar to spherical coordinates

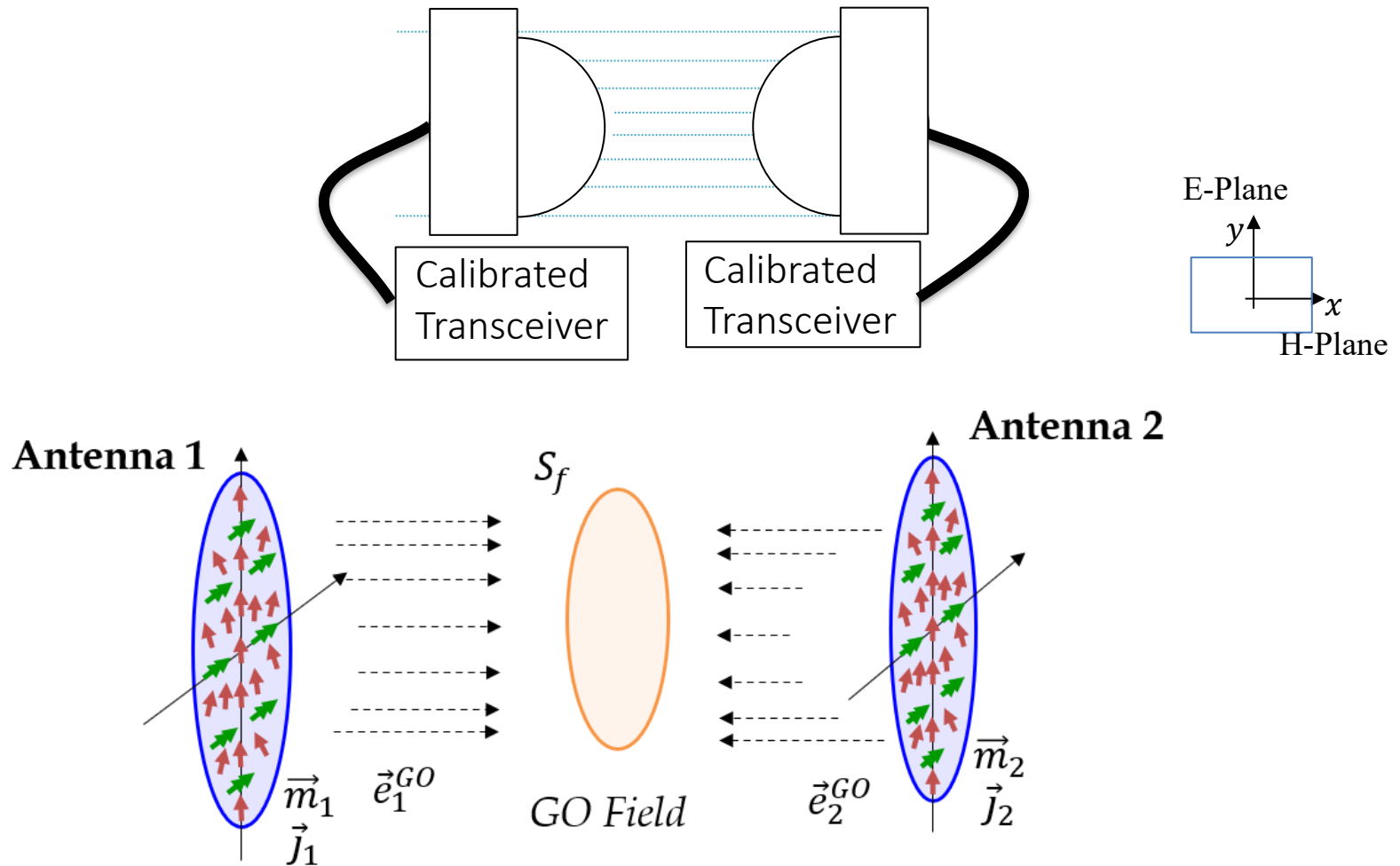


$$\text{For large f-numbers: } r' \approx r \rightarrow \sin \theta = \frac{x}{F}$$

2. Gain: For an electrically large DUT antenna and small probe aperture:

$$\frac{P_{rx}}{P_{tx}} = G_{LA} G_{Probe} \left(\frac{\lambda}{4\pi R} \right)^2$$

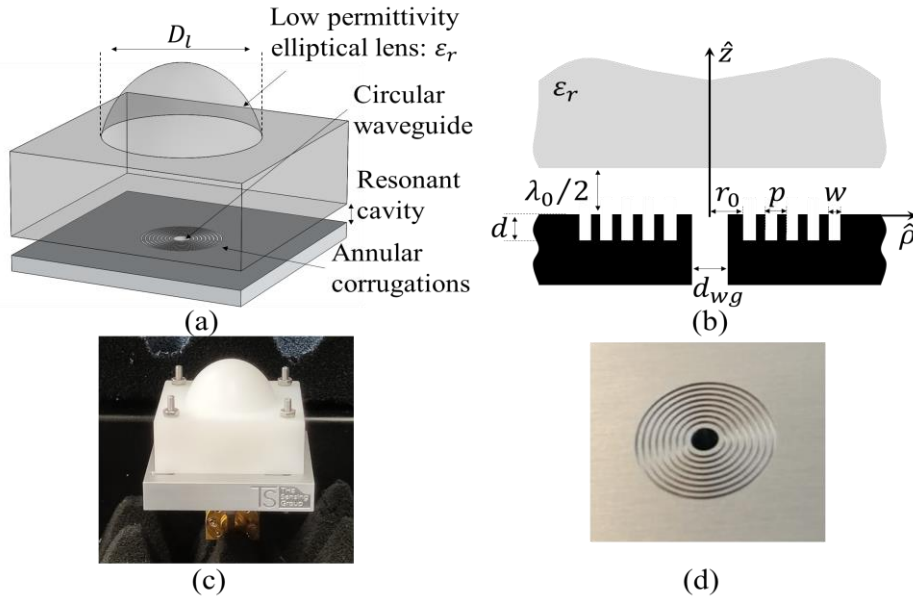
Lens to Lens Measurement



$$Cou = \frac{|V_{oc} I_0|^2}{16 P_{rad}^2 P_{rad}^1}$$

$$V_{oc} I_0 = \frac{2}{\zeta} \iint_{S_f} \vec{e}_1^{GO} \cdot \vec{e}_2^{GO} dS_f$$

Lens Antenna

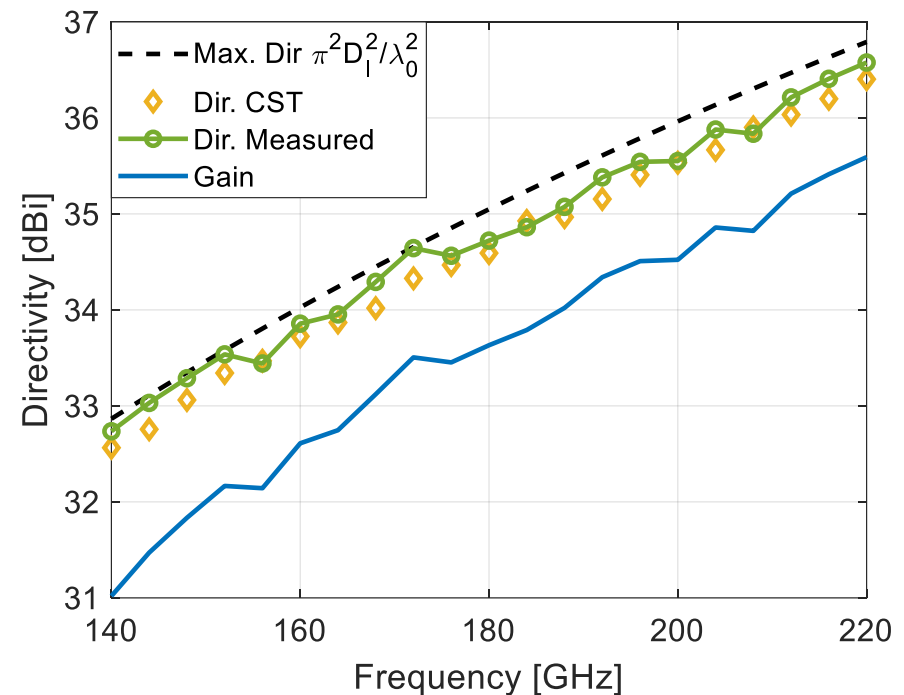
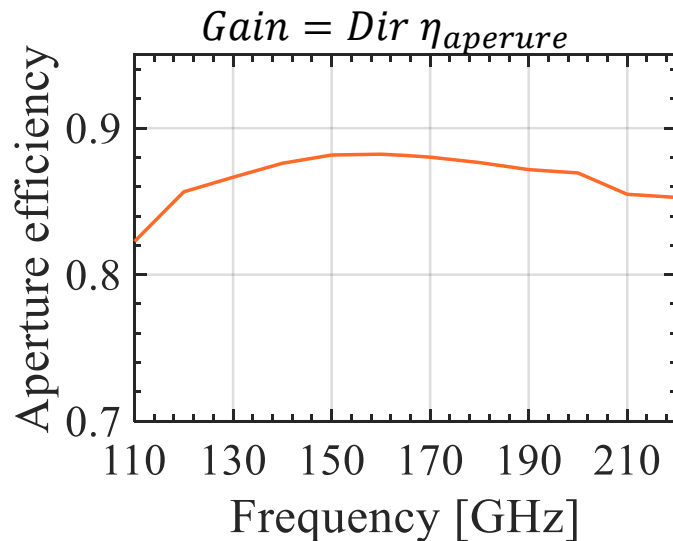


Center frequency $f_c = 165\text{GHz}$

Lens Diameter $D_L = 30\text{ mm}$

Lens subtended angle $\theta_{st} = 36.5^\circ$

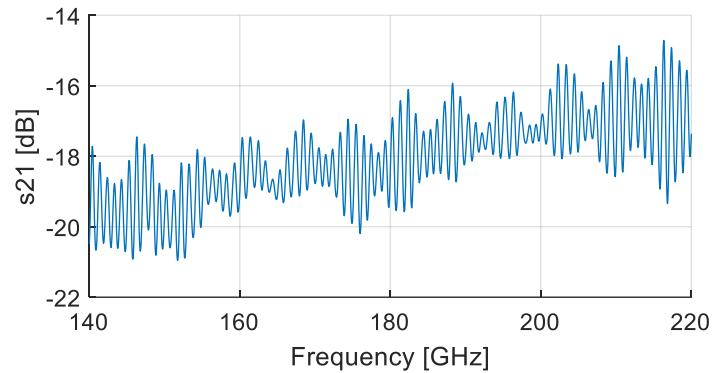
Phase Centre $\Delta z = -1.2\text{mm}$



Gain includes the Ohmic loss from the metal block and plastic

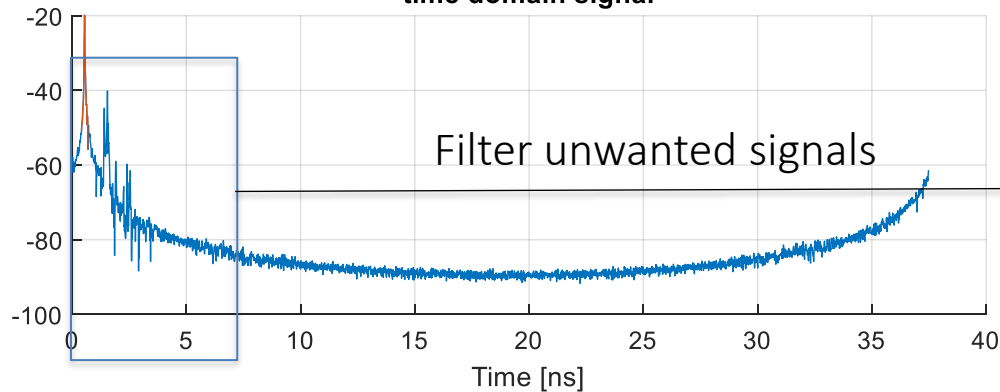
Time-gating

Measured Frequency Domain Spectrum

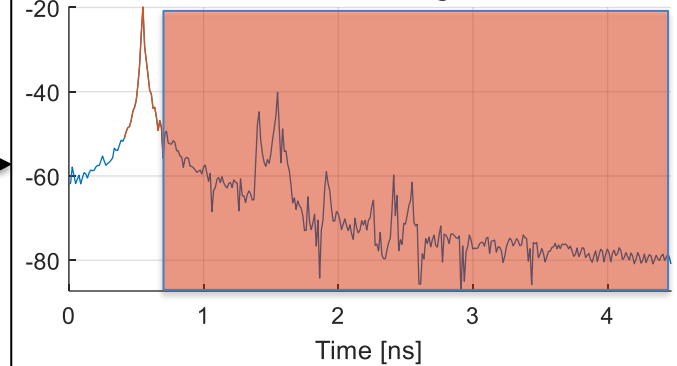


iFFT

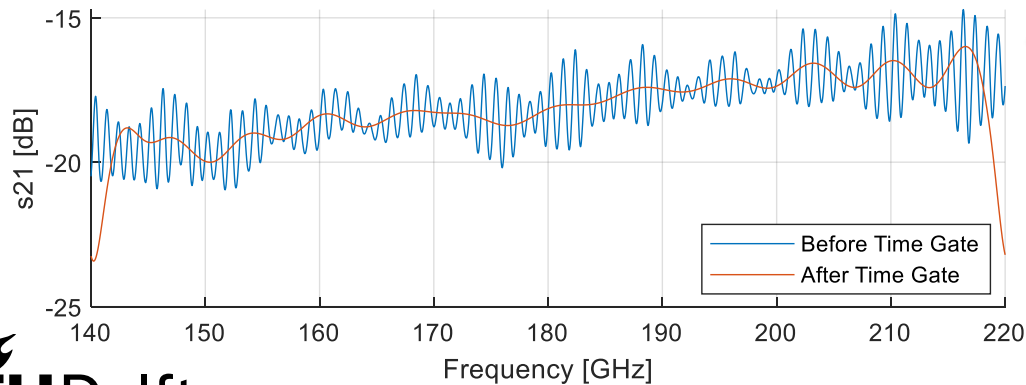
time domain signal



time domain signal

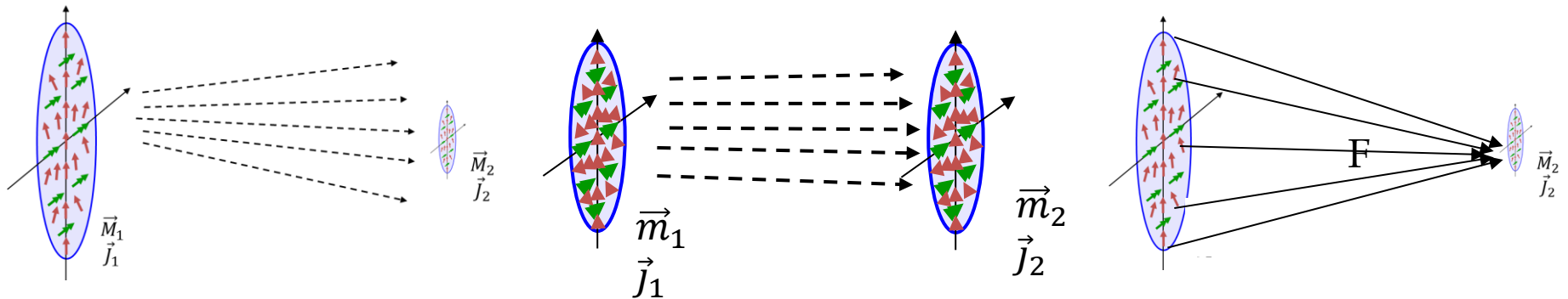


FFT



Report

- The radiation pattern of the horn and lens-antenna. Comparison with airy pattern.
- Directivity and gain vs frequency. Include directivity of ideal antenna (airy)
- Apply time-gating techniques to reduce the multiple reflections of the environment
- Compare coupling of the different setups.



Files in brightspace:

- Measurement data from the laboratory
- Previous 2D measurement scan in the near field of the lens antenna

